VOLV-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: Degrees to Radians: The Natural Angle

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is 360 an arbitrary number invented by the ancient Babylonians?'
- Ask: 'Why is radian measure called natural mathematics?'
- Ask: 'How does walking around a 1-meter radius track relate to radians?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students thinking pi radians means 3.14 degrees. Emphasize: pi radians is 180 DEGREES!

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"Convert 210° to exact radians using the Volvelle."
""

OFFICIAL SOLUTION KEY:

210° = 7 × 30° = 7pi/6 rad.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/unit-circle-volvelle/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution
within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

VOLV-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: Degrees to Radians: The Natural Angle

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why do mathematicians and engineers abandon 360 degrees and use radians instead? Because in a unit circle (r = 1), radians ARE the physical distance you walk ar

STEP-BY-STEP APPARATUS SETUP:

1. Hold the Unit Circle Volvelle in your hands.
2. Rotate the upper cursor rotor until the hairline aligns with 0° (Positive X-axis).
3. Look into the reading window at 0°: notice '0 rad' and coordinate (1, 0).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Precision Dual-Rotor Volvelle Slide Calculator with Exact Radical Windows
DATUM TARGET:	Radian Invariant: Arc Length s = r · theta -> When r = 1, Angle in Radians EXACTLY Equals Arc Length
CALCULATOR KEY:	[180] [->RAD] -> 3.14159... (pi radians)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Additional Math (A-Math), Ch. 8 (Radian Measure)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Precalculus, Ch. 4, Sec. 4.1 • CCSS.MATH.HSF.TF.A.1 • TEKS P.4A
GRADE LEVEL BAND:	Grades 9-11 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 4 ('Trigonometry, Vectors & Polar'), pp. 74-86
HARDWARE UNBOXING BOOKLET:	VOLV-01 Booklet: Page 2 (Polar Invariant) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

VOLV-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE RADIAN CONVERSION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Rotate the rotor to 90° (top of circle): Read the exact radical window: 'pi/2 rad'.

Step 2: Rotate to 180°: Read 'pi rad'. Notice: Halfway around a circle is exactly pi (3.14159...).

Step 3: Rotate to 270°: Read '3pi/2 rad'. Rotate full circle to 360°: Read '2pi rad' (6.28318...).

Step 4: Align to landmark angles: 30° = pi/6; 45° = pi/4; 60° = pi/3. Notice: 30° goes into 180° exactly 6 times!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Radian Invariant: Arc Length s = r · theta -> When r = 1, Angle in Radians EXACTLY Equals Arc Length

PHYSICAL APPARATUS MANIFEST:

Precision Dual-Rotor Volvelle Slide Calculator with Exact Ra
dical Windows

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

VOLV-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: FINDING SPECIAL ANGLES

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Align the rotor to 120°, 135°, and 150° in Quadrant II.

FORM A MASTER STATION TASKS (Complete All 3):

1. **What is 120° in exact radians?**
   Expected Physical Finding: 2pi/3 rad

2. **What is 135° in exact radians?**
   Expected Physical Finding: 3pi/4 rad

3. **What is 150° in exact radians?**
   Expected Physical Finding: 5pi/6 rad

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [180] [->RAD] -> 3.14159... (pi radians)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
180	X: 180	Angle in degrees
->RAD	X: 3.14159	Evaluate exact radian equivalent (pi rad)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

VOLV-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 4: Cartesian Projections: Cosine & Sine

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why can cosine and sine never be greater than 1 on a unit circle?'
- Ask: 'What happens to the shadows at 180°?'
- Ask: 'How does the Volvelle eliminate the need to memorize the unit circle table?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students mixing up X and Y. Cosine is ALWAYS horizontal X (like a shadow on the floor); Sine is ALWAYS vertica
l Y (like height on the wall).

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"What are the coordinates at 240° in Quadrant III?"
""

OFFICIAL SOLUTION KEY:

(−1/2, −sqrt(3)/2) — both coordinates are negative!

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
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FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK)

VOLV-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 1: Cartesian Projections: Cosine & Sine

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

What do cosine and sine actually measure? They are nothing more than the horizontal shadow (X) and vertical shadow (Y) of a rotating clock hand of length 1.

STEP-BY-STEP APPARATUS SETUP:

1. Align the Volvelle rotor to 30°.
2. Look at the coordinate window: notice (sqrt(3)/2, 1/2).
3. Inspect the physical grid: Notice the horizontal X coordinate reaches ~0.866 and vertical Y reaches exactly 0.500.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Dual-Rotor Volvelle with Exact Coordinate Readout Windows (cos theta, sin theta)
DATUM TARGET:	Coordinate Invariant: Any Point on Unit Circle P(theta) = (cos theta, sin theta)
CALCULATOR KEY:	[30] [COS] -> 0.8660 (sqrt(3)/2) | [30] [SIN] -> 0.5000 (1/2)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

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BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Precalculus, Ch. 4, Sec. 4.2 • CCSS.MATH.HSF.TF.A.2 • TEKS P.4B
GRADE LEVEL BAND:	Grades 10–12 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 4, pp. 86–94
HARDWARE UNBOXING BOOKLET:	VOLV-01 Booklet: Page 5 (Reading Coordinates) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

VOLV-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 2: LAB INVESTIGATION: THE 45° AND 60° INVERSIONS

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Rotate to 45°: Read coordinate (sqrt(2)/2, sqrt(2)/2) = (0.707, 0.707). Both shadows are equal!

Step 2: Rotate to 60°: Read coordinate (1/2, sqrt(3)/2) = (0.500, 0.866).

Step 3: Compare 30° and 60°: Notice the X and Y coordinates SWAPPED! 30° has X=sqrt(3)/2, Y=1/2; 60° has X=1/2, Y=sqrt(3)/2.

Step 4: Rotate to 90°: X shadow collapses to 0; Y shadow reaches maximum 1. (cos 90° = 0, sin 90° = 1).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

- ✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
- ✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
- ✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
- ✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:

Coordinate Invariant: Any Point on Unit Circle P(theta) = (cos theta, sin theta)

PHYSICAL APPARATUS MANIFEST:

Dual-Rotor Volvelle with Exact Coordinate Readout Windows (cos theta, sin theta)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

VOLV-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THE SIGN REVERSAL IN QUADRANT II

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Rotate the rotor to 150° in Quadrant II (top-left).

FORM A MASTER STATION TASKS (Complete All 3):

1. **What is the sign of the X coordinate (cosine) in Quadrant II?**
   Expected Physical Finding: **Negative (-sqrt(3)/2 ≈ -0.866)**
2. **What is the sign of the Y coordinate (sine) in Quadrant II?**
   Expected Physical Finding: **Positive (+1/2 = 0.500)**
3. **Why is cosine negative in Quadrant II?**
   Expected Physical Finding: **Because the point sits to the LEFT of the vertical Y-axis!**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [30] [COS] -> 0.8660 (sqrt(3)/2) | [30] [SIN] -> 0.5000 (1/2)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
30	X: 30	Angle 30 degrees
COS	X: 0.8660	Horizontal coordinate = sqrt(3)/2
30	Y: 0.8660 | X: 30	Angle
SIN	X: 0.5000	Vertical coordinate = 1/2

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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VOLV-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 4: The Pythagorean Invariant

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Can the sum of squares ever equal 1.05 or 0.98?'
- Ask: 'How does this identity protect GPS satellites from calculation drift?'
- Ask: 'If cos theta = 0.6, how can you find sin theta in 5 seconds without a calculator?' (sqrt(1 − 0.36) = 0.8!).

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students writing cos(theta)² instead of (cos theta)². Clarify that the notation cos² theta means you compute t
he cosine first, then square the result.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"If sin theta = 5/13 in Quadrant I, what is cos theta?"
""

OFFICIAL SOLUTION KEY:

cos theta = sqrt(1 − (5/13)²) = sqrt(144/169) = 12/13.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

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VOLV-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 1: The Pythagorean Invariant

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why is cos² theta + sin² theta = 1 the single most important identity in engineering? Because it is just the Pythagorean theorem (a² + b² = c²) written on a circ

STEP-BY-STEP APPARATUS SETUP:

1. Pick any angle on the Volvelle (e.g. 30°, 45°, 60°, or an odd angle like 37°).
2. Notice the triangle formed by the origin (0,0), the coordinate point (X, Y), and the X-axis.
3. The base is length cos theta; the height is length sin theta; the hypotenuse is the radius (r = 1).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Unit Circle Volvelle + SC-32 Scientific Hardware Calculator

DATUM TARGET: Pythagorean Theorem: In Right Triangle with Hypotenuse r=1: X² + Y² = r² -> cos² theta + sin² theta = 1

CALCULATOR KEY: [37] [COS] [x²] [37] [SIN] [x²] [+] -> 1.0000

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

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BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Precalculus, Ch. 5, Sec. 5.1 • CCSS.MATH.HSF.TF.C.8 • TEKS P.4D
GRADE LEVEL BAND:	Grades 10–12 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 4, pp. 94–102
HARDWARE UNBOXING BOOKLET:	VOLV-01 Booklet: Page 2 (Pythagorean Invariant) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
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3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

VOLV-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE SUM OF SQUARES AUDIT

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Test angle 30°: cos(30°) = sqrt(3)/2; sin(30°) = 1/2.

Step 2: Square both: (sqrt(3)/2)² = 3/4. (1/2)² = 1/4.

Step 3: Add them: 3/4 + 1/4 = 4/4 = 1.0000!

Step 4: Test angle 45°: (sqrt(2)/2)² + (sqrt(2)/2)² = 2/4 + 2/4 = 4/4 = 1.0000!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Pythagorean Theorem: In Right Triangle with Hypotenuse r=1: $X^2 + Y^2 = r^2 \rightarrow \cos^2\theta + \sin^2\theta = 1$

PHYSICAL APPARATUS MANIFEST:

Unit Circle Volvelle + SC-32 Scientific Hardware Calculator

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

VOLV-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: TESTING NON-STANDARD ANGLES

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Pick an arbitrary angle: 73.4° on your SC-32 calculator.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Calculate cos(73.4°).**
   Expected Physical Finding: **0.2857**
2. **Calculate sin(73.4°).**
   Expected Physical Finding: **0.9583**
3. **Sum their squares: (0.2857)² + (0.9583)².**
   Expected Physical Finding: **0.0816 + 0.9184 = 1.0000 (Exact!)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [37] [COS] [x²] [37] [SIN] [x²] [+] -> 1.0000

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
37	X: 37	Arbitrary angle
COS	X: 0.7986	cos(37°)
x²	X: 0.6378	cos²(37°)
37	Y: 0.6378 | X: 37	Angle
SIN	X: 0.6018	sin(37°)
x²	X: 0.3622	sin²(37°)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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VOLV-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 4: Tangent & The Infinite Horizon

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why is the tangent line called a tangent line?' (It touches the circle at exactly one point!).
- Ask: 'Can tangent be negative? Where?' (Quadrants II and IV, where slope is downhill).
- Ask: 'At what angle does tan theta equal -1?' (135° and 315°).

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students thinking tangent can't be greater than 1 because sine and cosine can't. Point out that slope can be a
s steep as a cliff!

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"Find an angle whose tangent is equal to its cosine. Can this happen?"
""

OFFICIAL SOLUTION KEY:

sin/cos = cos -> sin = cos² = 1 − sin² -> sin² + sin − 1 = 0 -> sin theta ≈ 0.618 (Golden Ratio angle ≈ 38.17°).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/unit-circle-volvelle/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution
within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

VOLV-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 1: Tangent & The Infinite Horizon

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does the tangent function shoot up to infinity at 90°? Because tangent is the slope of the angle line, and a vertical line has infinite slope.

STEP-BY-STEP APPARATUS SETUP:

1. Look at the vertical tangent ruler running along the right side of the Volvelle (at the x = 1 tangent line).
2. As you rotate the angle arm, the projection ray extends outward until it intersects this vertical line.
3. The height where it hits the line is the TANGENT of the angle.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Volvelle Slide Ruler with Exterior Tangent Projection Scale (at x = 1 line)
DATUM TARGET:	Tangent Invariant: tan theta = Slope of Radial Line = y / x = sin theta / cos theta
CALCULATOR KEY:	[89.9] [TAN] -> 572.95 | [90] [TAN] -> ERROR (Division by Zero)

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Additional Math, Ch. 9, Lesson 9.4 (Tangent Function)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Precalculus, Ch. 4, Sec. 4.3 • CCSS.MATH.HSF.TF.B.5 • TEKS P.4C
GRADE LEVEL BAND:	Grades 11–12 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 4, pp. 102–110
HARDWARE UNBOXING BOOKLET:	VOLV-01 Booklet: Page 6 (Tangent Tangency Line & Asymptote)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

VOLV-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE TANGENT ASCENT

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Rotate to 0°: Line is horizontal (slope = 0). Reading on tangent ruler: 0.00.

Step 2: Rotate to 45°: Slope is 1 (45° line rises 1 for every 1 run). Reading: exactly 1.000.

Step 3: Rotate to 60°: Slope steepens. Reading: sqrt(3) ≈ 1.732.

Step 4: Rotate toward 90°: Look at the ray: At 85°, it hits 11.43. At 89°, it hits 57.3. At 89.9°, it hits 573! At 90°, the line is parallel to the tangent ruler and NEVER touches it!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

- ✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
- ✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
- ✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
- ✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Tangent Invariant: tan theta = Slope of Radial Line = y / x = sin theta / cos theta

PHYSICAL APPARATUS MANIFEST:

Volvelle Slide Ruler with Exterior Tangent Projection Scale
(at x = 1 line)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

VOLV-01 • TENT 4/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: THE DIVISION BY ZERO ASYMPTOTE

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Explain why tan(90°) is undefined using sine and cosine.

FORM A MASTER STATION TASKS (Complete All 3):

1. **What is sin(90°)?**
   Expected Physical Finding: **1.000**
2. **What is cos(90°)?**
   Expected Physical Finding: **0.000**
3. **Compute tan(90°) = sin(90°) / cos(90°).**
   Expected Physical Finding: **1 / 0 = Undefined (Vertical Asymptote)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [89.9] [TAN] -> 572.95 | [90] [TAN] -> ERROR (Division by Zero)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
89.9	X: 89.9	Near vertical angle
TAN	X: 572.96	Tangent explodes to infinity
90	X: 90	Vertical line
TAN	X: NaN / ERROR	cos(90°) = 0 -> Division by zero asymptote

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

VOLV-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 4: The 4-Quadrant Sign Matrix

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does tangent end up positive in Quadrant III if both X and Y are negative?'
- Ask: 'How does finding the reference angle let you solve any trig problem on Earth using only Quadrant I?'
- Ask: 'What happens to cosine and sine on the boundary axes (0°, 90°, 180°, 270°)?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to remember signs by rote memorization. Just ask: 'Is the point to the right or left? Above or below the floor?'

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"If tan theta > 0 and sin theta < 0, in which quadrant must theta lie?"
""

OFFICIAL SOLUTION KEY:

Quadrant III (where tangent is positive and sine is negative).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/unit-circle-volvelle/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
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- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

VOLV-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 1: The 4-Quadrant Sign Matrix

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does every precalculus student learn the mnemonic 'All Students Take Calculus'? In this lab, rotate through all 4 quadrants to see the sign changes happen in

STEP-BY-STEP APPARATUS SETUP:

1. Rotate the Volvelle through all four quadrants.
2. Quadrant I (Top-Right, 0°–90°): X > 0, Y > 0.
3. Quadrant II (Top-Left, 90°–180°): X < 0, Y > 0.
4. Quadrant III (Bottom-Left, 180°–270°): X < 0, Y < 0.
5. Quadrant IV (Bottom-Right, 270°–360°): X > 0, Y < 0.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Dual-Rotor Volvelle with Quadrant Sign Indicators (QI: +/+, QII: −/+, QIII: −/−, QIV: +/−)

DATUM TARGET: ASTC Rule: QI (All Positive), QII (Sin Positive), QIII (Tan Positive), QIV (Cos Positive)

CALCULATOR KEY: [210] [SIN] -> −0.500 | [210] [COS] -> −0.866

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Additional Math, Ch. 9, Lesson 9.2 (Trig Signs in 4 Quadrants)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Precalculus, Ch. 4, Sec. 4.4 • CCSS.MATH.HSF.TF.A.2 • TEKS P.4B

GRADE LEVEL BAND: Grades 10–12 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 2: SC-32 Scientific Handbook, Ch. 4, pp. 110–118

HARDWARE UNBOXING BOOKLET: VOLV-01 Booklet: Page 5 (Quadrant Signs) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

VOLV-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 2: LAB INVESTIGATION: THE ASTC VERIFICATION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Quadrant I (ALL): cos, sin, and tan are all POSITIVE (+, +, +).

Step 2: Quadrant II (SIN): X is negative, Y is positive. Therefore, ONLY SINE is positive (+). (cos is −, tan is −).

Step 3: Quadrant III (TAN): X is negative, Y is negative. Negative / Negative = Positive! Therefore, ONLY TANGENT is positive (+).

Step 4: Quadrant IV (COS): X is positive, Y is negative. Therefore, ONLY COSINE is positive (+).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
ASTC Rule: QI (All Positive), QII (Sin Positive), QIII (Tan Positive), QIV (Cos Positive)

PHYSICAL APPARATUS MANIFEST:

Dual-Rotor Volvelle with Quadrant Sign Indicators (QI: +/+,
QII: −/+, QIII: −/−, QIV: +/−)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

VOLV-01 • TENT 5/6 | ADVANCED | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: REFERENCE ANGLES

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Every angle in QII, QIII, and QIV has a twin 'Reference Angle' in QI.

FORM A MASTER STATION TASKS (Complete All 3):

1. **What is the reference angle for 150° (Quadrant II)?**
   Expected Physical Finding: 180° − 150° = 30°

2. **What is the reference angle for 225° (Quadrant III)?**
   Expected Physical Finding: 225° − 180° = 45°

3. **What is the reference angle for 300° (Quadrant IV)?**
   Expected Physical Finding: 360° − 300° = 60°

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [210] [SIN] –> −0.500 | [210] [COS] –> −0.866

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
210	X: 210	Quadrant III angle
SIN	X: −0.500	Vertical component negative
210	Y: −0.500 | X: 210	Angle
COS	X: −0.866	Horizontal component negative

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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VOLV-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Physics Vectors: Force Resolution

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why can\'t you just add 40 N + 30 N to get 70 N?'
- Ask: 'How does the unit circle provide the direction vector for any magnitude in physics?'
- Ask: 'How does the SC-32 ->REC and ->POL key automate AC electrical engineering phasor math?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding vector magnitudes directly. Forces in different directions MUST be resolved into components be
fore adding!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"An aircraft flies at 200 mph at a bearing of 45° with a 20 mph tailwind from the South. What is its ground speed?"
""

OFFICIAL SOLUTION KEY:

Aircraft: (141.4, 141.4). Wind: (0, 20). Net: (141.4, 161.4). Ground speed = sqrt(141.4² + 161.4²) ≈ 214.6 mph.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/unit-circle-volvelle/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

VOLV-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Physics Vectors: Force Resolution

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

When a rocket launches at an angle or a crane pulls on a cable, the force acts diagonally. Engineers must resolve that diagonal pull into horizontal and vertica

STEP-BY-STEP APPARATUS SETUP:

1. Scenario: A cable pulls a crate with a force of 50 Newtons at an angle of 30° above the ground.
2. Set the Volvelle rotor to 30°.
3. Read the unit circle coordinates: (0.866, 0.500).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Unit Circle Volvelle + SC-32 Hardware Polar-to-Rectangular Engine
DATUM TARGET:	Vector Resolution: F_x = |F| · cos theta | F_y = |F| · sin theta
CALCULATOR KEY:	[50] [ENTER] [30] [->REC] -> X: 43.30, Y: 25.00

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Additional Math, Ch. 10 (Vectors in 2 Dimensions)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Secondary STEM & Advanced Inquiry Extension
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Precalculus, Ch. 6, Sec. 6.3 • CCSS.MATH.HSN.VM.A.3 • TEKS P.4H
GRADE LEVEL BAND:	Grades 11–12+ • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 4 ('Vectors & Phasors'), pp. 118–130
HARDWARE UNBOXING BOOKLET:	VOLV-01 Booklet: Page 7 (RPN Vector Polar/Rectangular) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

VOLV-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: SCALING THE UNIT VECTOR

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: The unit circle vector has length 1: F_unit = (cos 30°, sin 30°) = (0.866, 0.500).

Step 2: Scale the vector by 50 Newtons: Multiply both components by 50.

Step 3: Horizontal pull: F_x = 50 × cos(30°) = 50 × 0.8660 = 43.30 N.

Step 4: Vertical lift: F_y = 50 × sin(30°) = 50 × 0.5000 = 25.00 N.

Step 5: Execute on SC-32: [50] [ENTER] [30] [->REC]. Verify Register X reads 43.30 N and Register Y reads 25.00 N!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Vector Resolution: F_x = |F| · cos theta | F_y = |F| · sin theta

PHYSICAL APPARATUS MANIFEST:

Unit Circle Volvelle + SC-32 Hardware Polar-to-Rectangular E
ngine

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

--- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ---

VOLV-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: RESULTANT FORCE SUMMATION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

THE NON-ROUTINE PUZZLE:

Two tugboats pull a barge: Tug A pulls 40 N at 0° (due East). Tug B pulls 30 N at 90° (due North).

FORM A MASTER STATION TASKS (Complete All 3):

1. **Combine vectors: (40, 0) + (0, 30).**
   Expected Physical Finding: **(43, 30) -> Total Vector = (40, 30)**

2. **Calculate resultant magnitude on SC-32: [40] [ENTER] [30] [->POL].**
   Expected Physical Finding: **r = 50.00 N (3-4-5 Triangle!)**

3. **What is the compass heading angle theta?**
   Expected Physical Finding: **theta = 36.87°**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [50] [ENTER] [30] [->REC] -> X: 43.30, Y: 25.00

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
50	X: 50	Vector magnitude |F| = 50 Newtons
ENTER	Y: 50 | X: 50	Stack lift
30	Y: 50 | X: 30	Angle theta = 30 degrees
->REC	Y: 25.00 | X: 43.30	Convert Polar (r, theta) to Rectangular (Fx, Fy)

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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